

● HARD

● GEOMETRY AND TRIGONOMETRY

● ANSWER KEY

Geometry and Trigonometry

01

10 answers · Rationale-rich · Teach from doc

Q1**C**

C. The lengths of sides BC and EF

Choice C is correct. Since angles B and E each have the same measure and angles C and F each have the same measure, triangles ABC and DEF are similar, where side BC corresponds to side EF . To determine whether two similar triangles are congruent, it is sufficient to determine whether one pair of corresponding sides are congruent. Therefore, to determine whether triangles ABC and DEF are congruent, it is sufficient to determine whether sides BC and EF have equal length. Thus, the lengths of BC and EF are sufficient to determine whether triangle ABC is congruent to triangle DEF .

Choice A is incorrect and may result from conceptual errors.

Choice B is incorrect and may result from conceptual errors.

Choice D is incorrect. The given information is sufficient to determine that triangles ABC and DEF are similar, but not whether they are congruent.

Q2**A****A.** 149,796

Choice A is correct. The volume of a cube can be found by using the formula $V = s^3$, where V is the volume and s is the edge length of the cube. Therefore, the volume of the given cube is $V = 68^3$, or 314,432 cubic inches. The volume of a sphere can be found by using the formula $V = \frac{4}{3}\pi r^3$, where V is the volume and r is the radius of the sphere. Therefore, the volume of the given sphere is $V = \frac{4}{3}\pi 34^3$, or approximately 164,636 cubic inches. The volume of the space in the cube not taken up by the sphere is the difference between the volume of the cube and volume of the sphere. Subtracting the approximate volume of the sphere from the volume of the cube gives $314,432 - 164,636 = 149,796$ cubic inches.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Q3**A****A.** 0

Choice A is correct. The sine of a number t is the y -coordinate of the point arrived at by traveling a distance of t units counterclockwise around the unit circle from the starting point 1, 0. Since the unit circle has a circumference of 2π units, it follows that one full rotation around the circle is equal to a distance of 2π units. Therefore, a distance of 42π units around the circle from the starting point 1, 0 would result in exactly 21 full rotations, arriving back at the point 1, 0. So, $\sin 42\pi$ is equal to the y -coordinate of the point 1, 0, which is 0.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. This is the value of $\cos 42\pi$, not $\sin 42\pi$.

Q4**338**

The correct answer is 338. The tangent of an acute angle in a right triangle is the ratio of the length of the leg opposite the angle to the length of the leg adjacent to the angle. In triangle XYZ , it's given that angle Y is a right angle. Thus, $\bar{XY}$ is the leg opposite of angle Z and $\bar{YZ}$ is the leg adjacent to angle Z . It follows that $\tan Z = \frac{XY}{YZ}$. It's also given that the measure of angle Z is 33° and the length of $\bar{YZ}$ is 26 units. Substituting 33° for Z and 26 for YZ in the equation $\tan Z = \frac{XY}{YZ}$ yields $\tan 33^\circ = \frac{XY}{26}$. Multiplying each side of this equation by 26 yields $26 \tan 33^\circ = XY$. Therefore, the length of $\bar{XY}$ is $26 \tan 33^\circ$. The area of a triangle is half the product of the lengths of its legs. Since the length of $\bar{YZ}$ is 26 and the length of $\bar{XY}$ is $26 \tan 33^\circ$, it follows that the area of triangle XYZ is $\frac{1}{2} 26 26 \tan 33^\circ$ square units, or $338 \tan 33^\circ$ square units. It's given that the area, in square units, of triangle XYZ can be represented by the expression $k \tan 33^\circ$, where k is a constant. Therefore, 338 is the value of k .

Q5**54**

The correct answer is 54. The figure shown includes two triangles, triangle WQX and triangle YQZ , such that angle WQX and angle YQZ are vertical angles. It follows that angle WQX is congruent to angle YQZ . It's also given in the figure that the measures of angle W and angle Y are a° . Therefore angle W is congruent to angle Y . Since triangle WQX and triangle YQZ have two pairs of congruent angles, triangle WQX is similar to triangle YQZ by the angle-angle similarity postulate, where $\bar{YZ}$ corresponds to $\bar{WX}$, and $\bar{YQ}$ corresponds to $\bar{WQ}$. Since the lengths of corresponding sides in similar triangles are proportional, it follows that $\frac{YZ}{WX} = \frac{YQ}{WQ}$. It's given that $YQ = 63$, $WQ = 70$, and $WX = 60$. Substituting 63 for YQ , 70 for WQ , and 60 for WX in the equation $\frac{YZ}{WX} = \frac{YQ}{WQ}$ yields $\frac{YZ}{60} = \frac{63}{70}$. Multiplying each side of this equation by 60 yields $YZ = \frac{63}{70} 60$, or $YZ = 54$. Therefore, the length of $\bar{YZ}$ is 54.

Q6**B****B.** $318t + 80$

Choice B is correct. It's given that the base area of the right rectangular prism is $24t \text{ cm}^2$ and the length of the base is $\frac{8}{3} \text{ cm}$. Dividing the base area by the length yields the width: $\frac{24t}{\frac{8}{3}}$, or $\left(24t\right)\frac{3}{8}$, or $9t \text{ cm}$. It's also given that the height of the prism is 15 cm . A right rectangular prism has two rectangular bases and four rectangular lateral faces. The total area of the two bases is $2(24t)$, or $48t \text{ cm}^2$. The four lateral faces include two with dimensions $\frac{8}{3} \text{ cm}$ by 15 cm and two with dimensions $9t \text{ cm}$ by 15 cm . The total area of these four lateral faces is $2\frac{8}{3}(15) + 2(9t)(15) \text{ cm}^2$, which is equivalent to $80 + 270t \text{ cm}^2$. Adding this to the total area of the two bases yields $48t + 270t + 80$, or $318t + 80 \text{ cm}^2$. Thus, the expression, which represents the surface area, in cm^2 , of the right rectangular prism is $318t + 80$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. This is the volume, in cm^3 , not the surface area, in cm^2 .

Q7**A****A.** $-\sqrt{3}$

Choice A is correct. A trigonometric ratio can be found using the unit circle, that is, a circle with radius 1 unit. If a central angle of a unit circle in the xy -plane centered at the origin has its starting side on the positive x -axis and its terminal side intersects the circle at a point x, y , then the value of the tangent of the central angle is equal to the y -coordinate divided by the x -coordinate. There are 2π radians in a circle. Dividing $\frac{92\pi}{3}$ by 2π yields $\frac{92}{6}$, which is equivalent to $15 + \frac{2}{3}$. It follows that on the unit circle centered at the origin in the xy -plane, the angle $\frac{92\pi}{3}$ is the result of 15 revolutions from its starting side on the positive x -axis followed by a rotation through $\frac{2\pi}{3}$ radians. Therefore, the angles $\frac{92\pi}{3}$ and $\frac{2\pi}{3}$ are coterminal angles and $\tan\frac{92\pi}{3}$ is equal to $\tan\frac{2\pi}{3}$. Since $\frac{2\pi}{3}$ is greater than $\frac{\pi}{2}$ and less than π , it follows that the terminal side of the angle is in quadrant II and forms an angle of $\frac{\pi}{3}$, or 60° , with the negative x -axis. Therefore, the terminal side of the angle intersects the unit circle at the point $-\frac{1}{2}, \frac{\sqrt{3}}{2}$. It follows that the value of $\tan\frac{2\pi}{3}$ is $\frac{\frac{\sqrt{3}}{2}}{-\frac{1}{2}}$, which is equivalent to $-\sqrt{3}$. Therefore, the value of $\tan\frac{92\pi}{3}$ is $-\sqrt{3}$.

Choice B is incorrect. This is the value of $\frac{1}{\tan\frac{92\pi}{3}}$, not $\tan\frac{92\pi}{3}$.

Choice C is incorrect. This is the value of $\frac{1}{\tan\frac{\pi}{3}}$, not $\tan\frac{92\pi}{3}$.

Choice D is incorrect. This is the value of $\tan\frac{\pi}{3}$, not $\tan\frac{92\pi}{3}$.

Q8**C**c. $8\sqrt{3}$

Choice C is correct. It's given that the logo is in the shape of a trapezoid that consists of three congruent equilateral triangles, and that the perimeter of the trapezoid is 20 centimeters (cm). Since the perimeter of the trapezoid is the sum of the lengths of 5 of the sides of the triangles, the length of each side of an equilateral triangle is $\frac{20}{5} = 4$ cm. Dividing up one equilateral triangle into two right triangles yields a pair of congruent 30° - 60° - 90° triangles. The shorter leg of each right triangle is half the length of the side of an equilateral triangle, or 2 cm. Using the Pythagorean Theorem, $a^2 + b^2 = c^2$, the height of the equilateral triangle can be found. Substituting $a = 2$ and $c = 4$ and solving for b yields $\sqrt{4^2 - 2^2} = \sqrt{12} = 2\sqrt{3}$ cm. The area of one equilateral triangle is $\frac{1}{2}bh$, where $b = 4$ and $h = 2\sqrt{3}$. Therefore, the area of one equilateral triangle is $\frac{1}{2}(4)(2\sqrt{3}) = 4\sqrt{3}$ cm². The shaded area consists of two such triangles, so its area is $(2)(4)\sqrt{3} = 8\sqrt{3}$ cm².

Alternate approach: The area of a trapezoid can be found by evaluating the expression $\frac{1}{2}(b_1 + b_2)h$, where b_1 is the length of one base, b_2 is the length of the other base, and h is the height of the trapezoid. Substituting $b_1 = 8$, $b_2 = 4$, and $h = 2\sqrt{3}$ yields the expression $\frac{1}{2}(8 + 4)(2\sqrt{3})$, or $\frac{1}{2}(12)(2\sqrt{3})$, which gives an area of $12\sqrt{3}$ cm² for the trapezoid. Since two-thirds of the trapezoid is shaded, the area of the shaded region is $\frac{2}{3} \times 12\sqrt{3} = 8\sqrt{3}$.

Choice A is incorrect. This is the height of the trapezoid. Choice B is incorrect. This is the area of one of the equilateral triangles, not two. Choice D is incorrect and may result from using a height of 4 for each triangle rather than the height of $2\sqrt{3}$.

Q9**C**

C. 64

Choice C is correct. Since $RT = TU$, it follows that $\triangle RTU$ is an isosceles triangle with base RU. Therefore, $\angle TRU$ and $\angle TUR$ are the base angles of an isosceles triangle and are congruent. Let the measures of both $\angle TRU$ and $\angle TUR$ be t° . According to the triangle sum theorem, the sum of the measures of the three angles of a triangle is 180° . Therefore, $114^\circ + 2t^\circ = 180^\circ$, so $t = 33$.

Note that $\angle TUR$ is the same angle as $\angle SUV$. Thus, the measure of $\angle SUV$ is 33° . According to the triangle exterior angle theorem, an external angle of a triangle is equal to the sum of the opposite interior angles. Therefore, x° is equal to the sum of the measures of $\angle VSU$ and $\angle SUV$; that is, $31^\circ + 33^\circ = 64^\circ$. Thus, the value of x is 64.

Choice B is incorrect. This is the measure of $\angle STR$, but $\angle STR$ is not congruent to $\angle SVR$. Choices A and D are incorrect and may result from a calculation error.

Q10**2216**

The correct answer is 2,216. The surface area of a prism is the sum of the areas of all its faces. A right rectangular prism consists of six rectangular faces, where opposite faces are congruent. It's given that this prism has a length of 28 cm, a width of 15 cm, and a height of 16 cm. Thus, for this prism, there are two faces with area $28 \times 15 = 420 \text{ cm}^2$, two faces with area $28 \times 16 = 448 \text{ cm}^2$, and two faces with area $15 \times 16 = 240 \text{ cm}^2$. Therefore, the surface area, in cm^2 , of the right rectangular prism is $2(420) + 2(448) + 2(240)$, or 2,216.

Notes

Accept equivalent forms (e.g. $0.25 \equiv 1/4$). For grid-in items, decimal and fraction answers are both valid.

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